MTH636 – Game Theory

Lecture 14: Domination by Mixed Strategies & Nash's Theorem

1 Domination by Mixed Strategies

A pure strategy can be strictly dominated not only by another pure strategy, but also by a *mixed* strategy.

Example: Strict Domination by a Mixed Strategy

Consider the following game matrix:

$P_1 \setminus P_2$	L	C	R
T	(6, 2)	(0, 6)	(4, 4)
M	(2, 12)	(4, 3)	(2, 5)
B	(0, 6)	(10, 0)	(2, 2)

Observe that the pure strategy M for Player 1 is not strictly dominated by either T or B alone. However, consider the mixed strategy $\sigma_1 = [\frac{1}{2}(T), 0(M), \frac{1}{2}(B)]$. Let's calculate the expected payoff of σ_1 against each of Player 2's pure strategies:

- Against L: $U_1(\sigma_1, L) = \frac{1}{2}(6) + \frac{1}{2}(0) = 3 > 2 = U_1(M, L)$
- Against C: $U_1(\sigma_1, C) = \frac{1}{2}(0) + \frac{1}{2}(10) = 5 > 4 = U_1(M, C)$
- Against R: $U_1(\sigma_1, R) = \frac{1}{2}(4) + \frac{1}{2}(2) = 3 > 2 = U_1(M, R)$

Since $U_1(\sigma_1, s_2) > U_1(M, s_2)$ for all $s_2 \in S_2$, we have:

$$U_1(\sigma_1, \sigma_2) > U_1(M, \sigma_2) \quad \forall \sigma_2 \in \Sigma_2$$

Thus, M is strictly dominated by the mixed strategy σ_1.

Definition: Strict Domination (General Definition)

A strategy σ_i of player i **strictly dominates** a pure strategy s_i of player i if:

$$U_i(\sigma_i, \sigma_{-i}) > U_i(s_i, \sigma_{-i}) \quad \forall \sigma_{-i} \in \Sigma_{-i}$$

Equivalently, it suffices to check against all pure strategies of the opponent:

$$U_i(\sigma_i, s_{-i}) > U_i(s_i, s_{-i}) \quad \forall s_{-i} \in S_{-i}$$

Theorem: Strictly Dominated Strategies in Nash Equilibrium

Suppose for player i, a strategy σ_i strictly dominates a pure strategy s_i of player i. Then s_i **does not get a positive probability** in any Nash Equilibrium of the game.

Proof. Suppose $\overline{\sigma}$ is a mixed strategy profile and $\overline{\sigma}_i(s_i) > 0$. We will show that there exists a mixed strategy $\overline{\overline{\sigma}}_i$ such that $\overline{\overline{\sigma}}_i$ strictly dominates $\overline{\sigma}_i$ against any opponent strategy

profile, meaning $\overline{\sigma}$ cannot be a best response and thus cannot be part of an NE.

Construct the new strategy $\overline{\overline{\sigma}}_i$ by taking the probability mass assigned to the dominated strategy s_i and reassigning it proportionally to the components of the dominating strategy σ_i. Specifically, for any pure strategy $t_i \in S_i$:

$$\overline{\overline{\sigma}}_i(t_i) = \begin{cases} \overline{\sigma}_i(s_i) \cdot \sigma_i(s_i) & \text{if } t_i = s_i \\ \overline{\sigma}_i(t_i) + \overline{\sigma}_i(s_i)\sigma_i(t_i) & \text{if } t_i \neq s_i \end{cases}$$

By calculating the expected utility, we can sum the terms:

$$\begin{aligned} U_i(\overline{\overline{\sigma}}_i, \sigma_{-i}) &= \sum_{t_i \in S_i} \overline{\overline{\sigma}}_i(t_i) U_i(t_i, \sigma_{-i}) \\ &= \sum_{t_i \neq s_i} \left[\overline{\sigma}_i(t_i) + \overline{\sigma}_i(s_i)\sigma_i(t_i)\right] U_i(t_i, \sigma_{-i}) + \overline{\sigma}_i(s_i)\sigma_i(s_i) U_i(s_i, \sigma_{-i}) \\ &= \sum_{t_i \neq s_i} \overline{\sigma}_i(t_i) U_i(t_i, \sigma_{-i}) + \overline{\sigma}_i(s_i) \left(\sum_{t_i \neq s_i} \sigma_i(t_i) U_i(t_i, \sigma_{-i}) + \sigma_i(s_i) U_i(s_i, \sigma_{-i}) \right) \\ &= \sum_{t_i \neq s_i} \overline{\sigma}_i(t_i) U_i(t_i, \sigma_{-i}) + \overline{\sigma}_i(s_i) U_i(\sigma_i, \sigma_{-i}) \end{aligned}$$

Since σ_i strictly dominates s_i, we have $U_i(\sigma_i, \sigma_{-i}) > U_i(s_i, \sigma_{-i})$. Substituting this inequality gives:

$$\begin{aligned} U_i(\overline{\overline{\sigma}}_i, \sigma_{-i}) &> \sum_{t_i \neq s_i} \overline{\sigma}_i(t_i) U_i(t_i, \sigma_{-i}) + \overline{\sigma}_i(s_i) U_i(s_i, \sigma_{-i}) \\ &= U_i(\overline{\sigma}_i, \sigma_{-i}) \end{aligned}$$

Thus, $\overline{\overline{\sigma}}_i$ yields a strictly higher payoff than $\overline{\sigma}_i$, meaning $\overline{\sigma}_i$ cannot be a best response, and thus cannot be part of an NE. □

2 Proof of Nash's Theorem

Theorem: Nash's Existence Theorem

Suppose $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ is a strategic form game where all the players have finitely many pure strategies. Then G has at least one **mixed strategy Nash Equilibrium**.

To prove this, we rely on Brouwer's Fixed Point Theorem.

Theorem: Brouwer's Fixed Point Theorem

Let X be a convex and compact subset of $\mathbb{R}^d$ for some $d \in \mathbb{N}$, and let $f : X \to X$ be a continuous function. Then f has a fixed point, i.e., $\exists x \in X$ such that $f(x) = x$.

Proof of Nash's Theorem:
Let $m_i = |S_i|$. The space of mixed strategies for player i is the simplex:

$$\Sigma_i = \left\{ (x_1, \ldots, x_{m_i}) \mid x_j \geq 0 \ \forall j, \ \sum_{j=1}^{m_i} x_j = 1 \right\}$$

Claim 1: Σ_i is a convex and compact subset of $\mathbb{R}^{m_i}$.
Claim 2: Suppose A and B are convex and compact subsets of $\mathbb{R}^{d_1}$ and $\mathbb{R}^{d_2}$ respectively. Then $A \times B$ is a convex and compact subset of $\mathbb{R}^{d_1+d_2}$.
From Claims 1 and 2, the set of all mixed strategy profiles $\Sigma = \Sigma_1 \times \Sigma_2 \times \cdots \times \Sigma_n$ is a convex and compact subset of $\mathbb{R}^{\sum m_i}$.

We will construct a function $f : \Sigma \to \Sigma$ such that:

(i) f is continuous.

(ii) Every fixed point of f is a Nash Equilibrium of G.

By Brouwer's Fixed Point Theorem, such a function must have a fixed point, which will imply the existence of a Nash Equilibrium.

For a strategy profile $\sigma \in \Sigma$, let $\delta_i^j(\sigma)$ measure the "gain" a player gets by deviating to their j-th pure strategy s_i^j:

$$\delta_i^j(\sigma) = \max\left\{0,\ U_i(s_i^j, \sigma_{-i}) - U_i(\sigma)\right\}$$

Claim 3: δ_i^j is a continuous function of σ.
Proof of Claim 3: The expected utility $U_i(\sigma)$ is a multilinear function of the probabilities in σ, hence it is continuous. $U_i(s_i^j, \sigma_{-i})$ is similarly multilinear in σ_{-i} and thus continuous in σ. Since the difference of continuous functions is continuous, and the $\max(0, \cdot)$ function is continuous, $\delta_i^j(\sigma)$ is continuous.

(The full construction of f continues by using these δ_i^j values to strictly increase the probabilities of pure strategies that yield better-than-average payoffs, normalized so they sum to 1).